Bilkent University

Department of Electrical and Electronics Engineering

Lab 3 Report: Combinational Logic Circuit

Purpose

The primary objective of this laboratory session is to construct a combinational logic circuit on a breadboard. By integrating a 4-bit counter with logic gates and LEDs, the experiment aims to demonstrate the practical behavior of digital outputs in a physical hardware setting.

Design Specifications

For this experiment, I designed and built a combinational logic circuit on a breadboard. I utilized a 74HC163 counter to generate four input signals (Q0, Q1, Q2, and Q3), which continuously cycle from 0000 to 1111. Rather than producing a single combined output, I incorporated a single SN74HC32N Quad 2-input OR gate IC to evaluate two distinct logical functions: F1 = Q0 + Q1 and F2 = Q2 + Q3. The first function, F1, takes the two least significant bits (Q0 and Q1) as inputs, meaning it will output a logic HIGH (1) whenever either Q0 or Q1 is active. Similarly, the second function, F2, takes the two most significant bits (Q2 and Q3) as inputs, resulting in a HIGH output when either Q2 or Q3 is active. To visually monitor the circuit's operation, I wired the outputs of these specific gates to two separate LEDs. To protect these LEDs from excessive current that could potentially damage or burn them out, I incorporated 1000 Ω current-limiting resistors into the design. Finally, I established all power, ground, and clock connections based on the specifications provided in the respective component datasheets.

Methodology

I began the assembly by mounting the 74HC163 counter onto the breadboard. Following its pinout diagram, I tied the VCC, MR, CEP, CET, and PE terminals to a +5V source provided by a DC power supply, and I properly grounded the GND pin. To drive the counter, I applied a square wave from a signal generator to the clock (CP) pin. This square wave acts as the primary timing signal, dictating the specific speed at which the counter increments through its 16-state binary sequence.

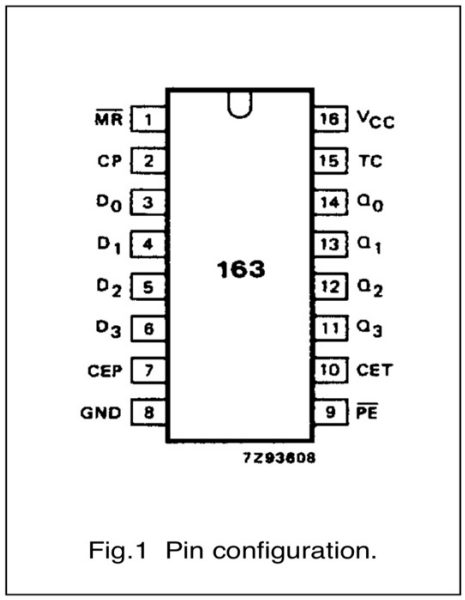


Figure 1: Pin Configuration for 4-bit Counter

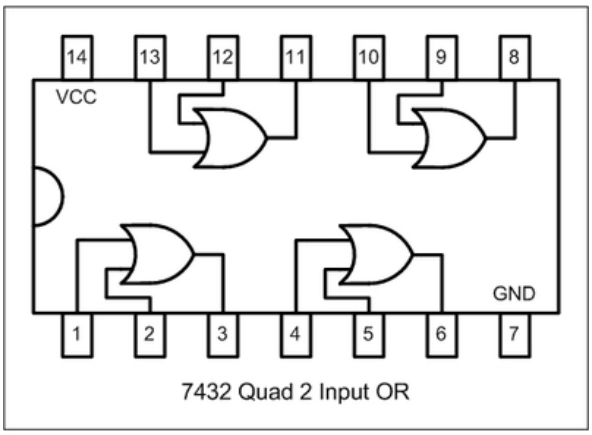


Figure 2: Schematic for 2-input OR Gate

Next, I routed the counter's outputs to the inputs of the SN74HC32N OR gate. I used two of the four internal OR gates for this setup. Specifically, I wired the Q0 and Q1 outputs from the counter to the 1A and 1B inputs of the first internal OR gate. I then wired the Q2 and Q3 outputs to the 2A and 2B inputs of the second internal OR gate. The respective outputs of these discrete gates, 1Y and 2Y, correspond directly to my F1 and F2 logic functions. I then linked the resulting outputs, F1 and F2, to indicator LEDs. Before connecting the gate outputs directly to the LEDs, I placed a 1000 Ω resistor in series with the anode of each LED, while the cathodes were tied to the common ground rail. This essential step ensures the current remains within safe operating limits for the discrete components. Additionally, I connected an oscilloscope to the circuit to confirm the accuracy of the generated clock signals.

Q3	Q2	Q1	Q0	F1(Q0+Q1)	F2(Q2+Q3)
0	0	0	0	0	0
0	0	0	1	1	0
0	0	1	0	1	0
0	0	1	1	1	0
0	1	0	0	0	1
0	1	0	1	1	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	0	1
1	0	0	1	1	1
1	0	1	0	1	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	0	1	1	1
1	1	1	0	1	1
1	1	1	1	1	1

Figure 3: Output Function of the Circuit

Results

During the experiment, I observed that the counter's outputs successfully demonstrated a standard binary progression. The least significant bit (Q0) alternated with every clock pulse, while the subsequent bits (Q1, Q2, and Q3) switched at progressively halved frequencies. The theoretical states for this sequence are detailed in the truth table below.

My oscilloscope measurements confirmed that the waveforms matched the anticipated counting pattern, indicating proper circuit functionality.

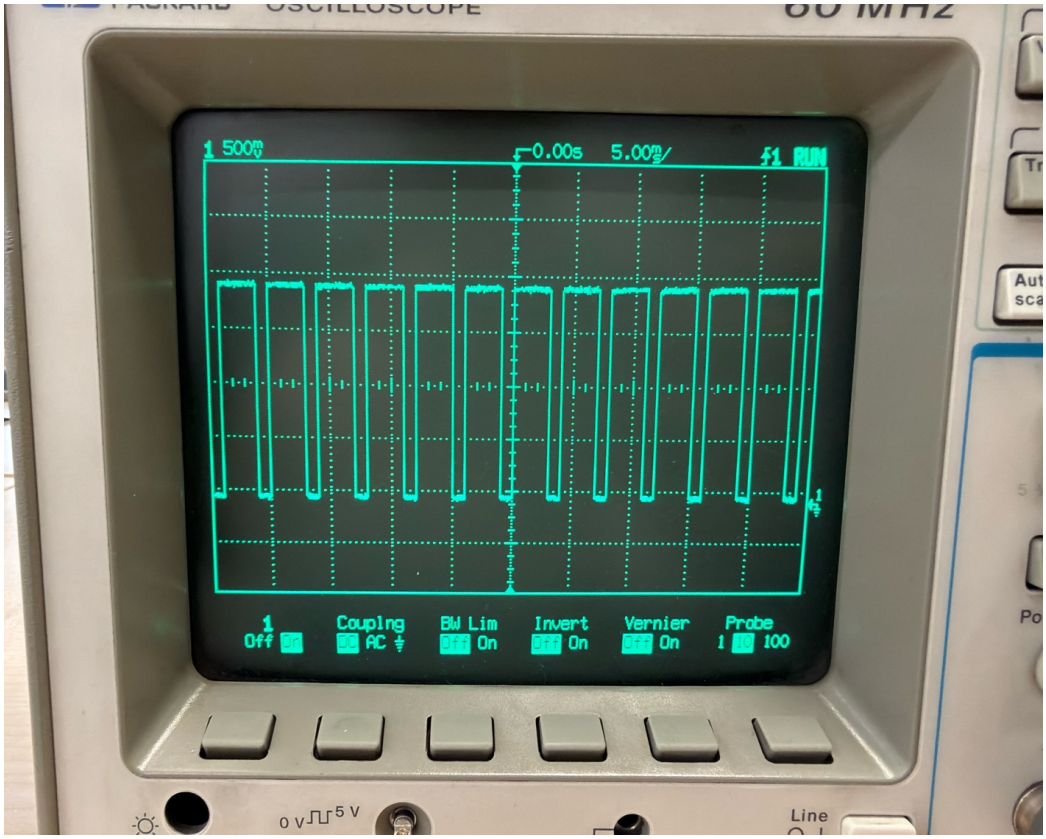


Figure 4: Oscilloscope Waveform

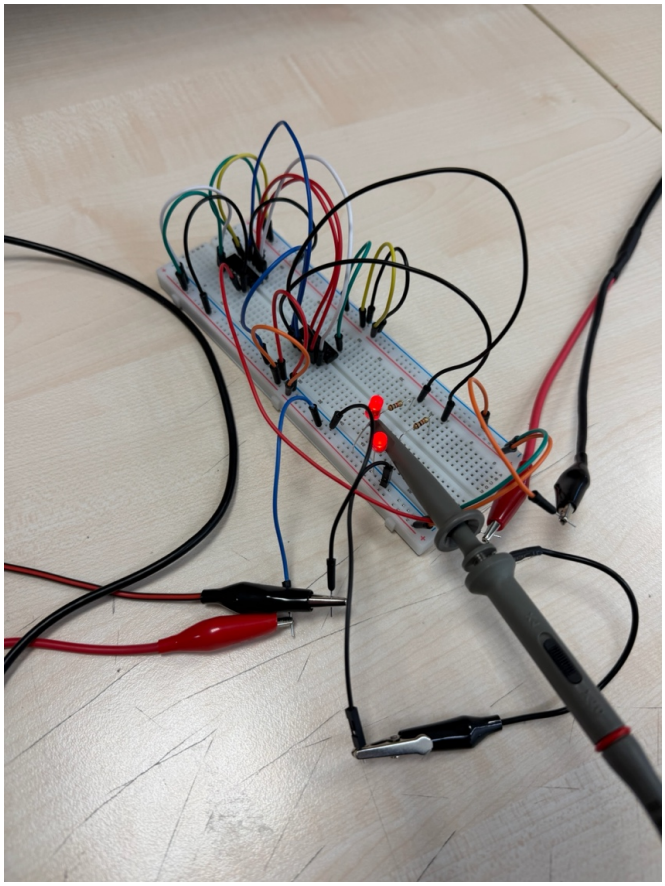


Figure 5: Circuit Connected to the Oscilloscope

Furthermore, by visually inspecting the LEDs throughout the clock cycles, I verified that the physical outputs precisely aligned with the F1 and F2 values predicted in the truth table. For instance, the LED corresponding to F1 blinked on and off rapidly due to the frequent state changes of the lower-order bits (Q0 and Q1), remaining off only when both were 0. Conversely, the LED corresponding to F2 remained off for the first four clock cycles and then stayed continuously illuminated for the remainder of the sequence, perfectly reflecting the steady behavior of the higher-order bits (Q2 and Q3).

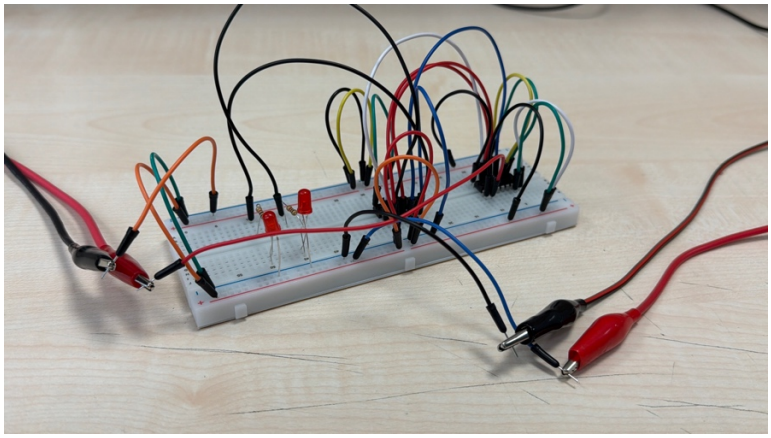


Figure 6: The Setup of the Circuit

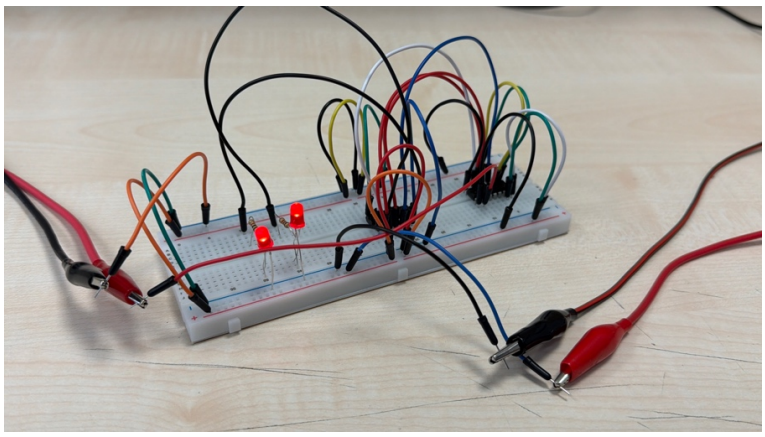


Figure 7: Outputs: F1 = 1, F2 = 1

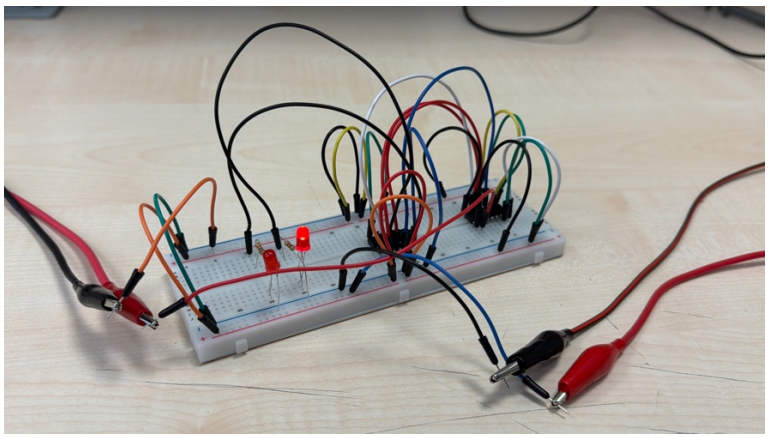


Figure 8: Outputs: F1 = 0, F2 = 1

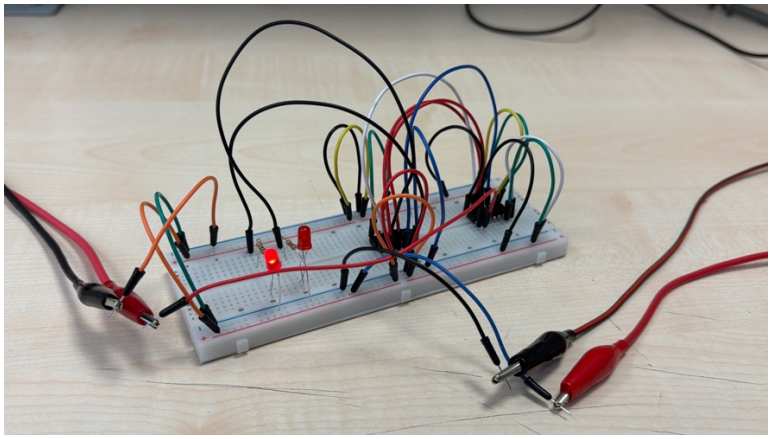


Figure 9: Outputs: F1 = 1, F2 = 0

Conclusion

Throughout this laboratory exercise, I successfully constructed and tested a combinational logic circuit using a 74HC163 4-bit counter and an SN74HC32N OR gate. The exact alignment between the theoretical truth table and the physical LED responses confirmed the accuracy of my implementation. This experiment offered me valuable hands-on experience in assembling digital hardware, interpreting component datasheets, calculating and implementing appropriate current-limiting resistors, and translating logic diagrams into working breadboard circuits. Moreover, utilizing an oscilloscope to monitor signals and troubleshoot connections significantly enhanced my practical understanding of waveform analysis in physical electronic systems.

References

74_HC_163.pdf

LogicGates.pdf

Texas Instruments, "SN74HC32 Quadruple 2-Input Positive-OR Gates Datasheet," available at: https://www.ti.com/lit/ds/symlink/sn74hc32.pdf

Texas Instruments, "SN74HC163 4-Bit Synchronous Binary Counters Datasheet," available at: https://www.ti.com/lit/ds/symlink/sn74hc163.pdf

Electronics Tutorials, "Combinational Logic Circuits," available at: https://www.electronics-tutorials.ws/combination/comb_1.html